Data Visualization with Tableau

Complete Notes: all class handouts combined and explained in simple language

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Part A: Introduction to Data Visualization

1. Why Visualize Data?

Data visualization means showing data as pictures such as charts, graphs, maps and dashboards. A picture is understood much faster than a long table of numbers, so it helps people understand information quickly and make better decisions.

Why is data visualization important?

1. Makes complex data easy to understand.
2. Shows trends and patterns quickly.
3. Highlights relationships between variables.
4. Detects outliers and anomalies (unusual values).
5. Supports faster and better decision-making.
6. Makes it easy to compare different datasets.
7. Improves communication of information.
8. Helps monitor business performance using dashboards.
9. Makes reports attractive and interactive.
10. Helps in prediction and forecasting from past (historical) data.

Benefits

- Better understanding of large datasets.
- Faster analysis than reading tables.
- Errors and inconsistencies are easy to spot.
- Better business intelligence and strategic planning.
- Better presentation of research findings.

Common types of visualization

Chart	Used for
Bar chart	Comparing different categories
Line chart	Showing trends over time
Pie chart	Showing proportions / percentages
Histogram	Showing frequency distribution
Scatter plot	Showing the relationship between two variables
Heat map	Showing intensity of values using colour
Area chart	Showing cumulative trends
Box plot	Showing spread of data and outliers

Applications

Business analytics, healthcare, finance, education, scientific research, marketing, government statistics and weather forecasting.

Example

Month	Jan	Feb	Mar	Apr
Sales	120	150	180	170

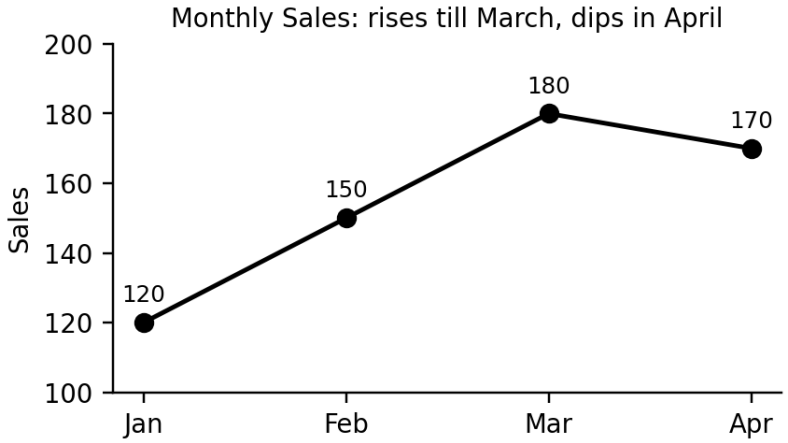


Figure: The same data as a line chart; the trend is visible at a glance

From the line chart we see immediately that sales increased from January to March and dropped slightly in April.

Remember: *A chart is understood faster than a table of numbers.*

2. History of Data Visualization

Data visualization is the practice of showing information graphically so data is easier to understand. Its history goes back thousands of years, from simple maps to today's interactive dashboards.

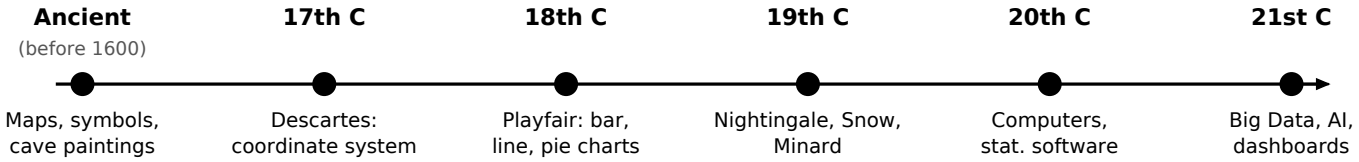


Figure: Timeline of data visualization

1. Ancient period (before 1600)

- Early humans used cave paintings and symbols to record information.
- Egyptians and Babylonians made maps for land measurement and navigation.
- Greek scholars used geometric diagrams to explain mathematics.

2. 17th century

- Scientific discoveries increased the need to show data visually.
- Astronomers and mathematicians started using graphs and coordinate systems.
- **René Descartes** introduced the **Cartesian coordinate system** (x-axis and y-axis), the foundation of modern graphs.

3. 18th century

- Modern statistical graphics began.
- **William Playfair** introduced the **bar chart (1786)**, the **line chart** and the **pie chart (1801)**.
- These charts became standard tools for comparing and analysing data.

4. 19th century

- Visualization spread into science, economics and public health.

- **Florence Nightingale** used the **Coxcomb (polar area) chart** to show causes of soldiers' deaths in the Crimean War (most deaths were from preventable disease, not wounds).
- **John Snow** plotted cholera deaths on a **map of London** and found the source of the outbreak (a water pump). This proved how useful visualization is in epidemiology.
- **Charles Joseph Minard** drew a famous **flow map of Napoleon's 1812 Russian campaign**, regarded as one of the greatest statistical graphics ever made.

5. 20th century

- Computers changed data visualization.
- Statistical software could create charts automatically.
- Businesses adopted dashboards and reporting systems.
- Interactive visualization became possible.

6. 21st century (modern era)

- Big Data, Artificial Intelligence and Machine Learning increased the demand for advanced visuals.
- Interactive dashboards allow real-time analysis.
- Popular tools: **Tableau, Microsoft Power BI, Google Looker Studio, Microsoft Excel, Python (Matplotlib, Seaborn, Plotly), R (ggplot2)**.

Timeline summary

Period	Major development
Ancient	Maps, symbols and diagrams
17th century	Cartesian coordinate system (Descartes)
18th century	Bar, line and pie charts (William Playfair)
19th century	Nightingale's Coxcomb chart, John Snow's cholera map, Minard's flow map
20th century	Computer-based charts and statistical software
21st century	Interactive dashboards, Big Data visualization, AI-powered analytics

Importance of this history

- Improved understanding of complex data.
- Made evidence-based decision-making possible.
- Advanced scientific and medical research.
- Supported business intelligence and analytics.
- Made information accessible to a wider audience.

Remember: *Playfair gave us the charts, Descartes gave us the axes, Nightingale and Snow proved charts can save lives.*

3. Who Uses Data Visualization?

Data visualization is used by many professionals and organizations to analyse data, communicate insights and support decision-making.

User	How they use it
Business Analysts	Analyse sales, revenue and customer behaviour; create dashboards and business reports
Data Scientists	Explore datasets, find patterns, present machine-learning results and predictions
Data Analysts	Clean, analyse and visualize data; make reports for decision-makers
Marketing Professionals	Track campaign performance; analyse customer demographics and market trends
Financial Analysts	Monitor stock prices and financial performance; analyse budgets, profits, investments
Healthcare Professionals	Visualize patient records and disease trends; monitor hospital and public-health data
Government Agencies	Present census, economic and crime statistics; support policy planning
Researchers and Scientists	Display experimental results; find trends and relationships in research data
Educators and Students	Present research and projects; understand complex concepts
Engineers	Monitor system performance; analyse sensor data and manufacturing processes
IT and Cybersecurity Teams	Track network traffic and system performance; detect security threats and unusual activity
Media and Journalists	Create infographics and interactive charts; explain data to the public

Remember: *Almost every field that has data uses visualization.*

Part B: Getting Data into Tableau

4. Getting Data with Tableau

Getting data is the **first step** in Tableau. Tableau lets you connect to data from many sources, clean it and prepare it for visualization.

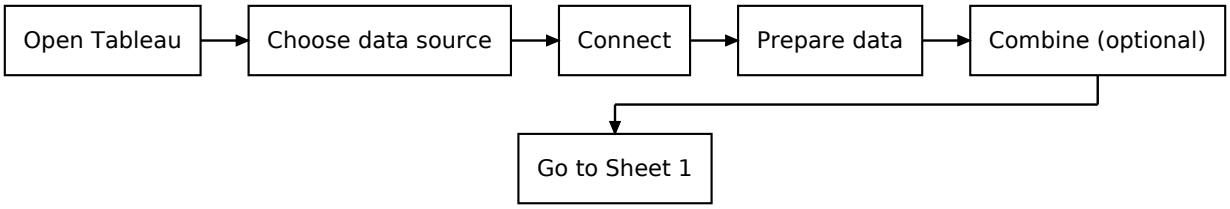


Figure: Steps to get data into Tableau

Steps to get data

1. **Open Tableau**: launch Tableau Desktop. The **Connect pane** appears on the start page.
2. **Choose a data source**: Excel, CSV (text file), PDF, SQL Server, MySQL, Oracle, PostgreSQL, Google Sheets, Microsoft Access, or cloud databases (Google BigQuery, Snowflake, Amazon Redshift).
3. **Connect to the data**: select the source, browse and open the file (or enter database details). Tableau loads it into the **Data Source page**.
4. **Prepare the data**: rename fields, change data types (text, number, date, Boolean), filter unwanted records, hide columns, clean missing or wrong values.
5. **Combine data (optional)**: **Joins** combine tables using a common field. **Relationships** connect related tables but keep them separate. **Unions** add rows from tables that have the same structure.
6. **Go to a worksheet**: click **Sheet 1** and drag fields to Rows, Columns, Marks or Filters.

Types of data connections

Type	Examples
File-based	Excel, CSV, JSON, PDF (with preprocessing)
Database	MySQL, SQL Server, Oracle, PostgreSQL
Cloud	Google Sheets, Google BigQuery, Snowflake, Amazon Redshift

Connection modes

- **Live connection**: connects directly to the original data and shows the latest data. Best for data that changes often.
- **Extract connection**: makes a local copy of the data. Faster for large data and can be refreshed when needed.

Advantages

- Connects to many data sources.
- Supports both live and extracted data.
- Easy drag-and-drop interface.
- Handles large datasets efficiently.
- Has built-in data preparation features.

5. Tableau Data Source

A **Tableau data source** is the **connection between Tableau and the data** you want to analyse. It defines **where** the data comes from, **how** it is connected and how Tableau **accesses** it.

Types of data sources

Type	Examples
Spreadsheet files	Microsoft Excel, CSV, text files
Databases	MySQL, SQL Server, Oracle, PostgreSQL
Cloud databases	Google BigQuery, Amazon Redshift, Snowflake
Cloud applications	Salesforce, Google Analytics
Web data	Web Data Connector, JSON, PDF
Statistical files	SAS, SPSS, R data
Big Data sources	Hadoop, Apache Hive, Spark

Connection types

- **Live**: connects directly to the original source, shows the latest data, needs continuous access to the source.
- **Extract**: makes a local snapshot, gives faster queries, works offline, must be refreshed from time to time.

Components of a data source

Component	Meaning
Connection	Links Tableau to the data
Tables	Data tables brought in from the source
Joins	Combine related tables
Relationships	Connect tables but keep each table's own structure
Data types	Number, String, Date, Boolean, Geographic
Metadata	Field names, aliases and descriptions

Common operations

Connect to data, add/remove tables, create joins and relationships, rename fields, change data types, create calculated fields, filter data, create extracts, refresh data.

Advantages

- Many data sources can be used in one workbook.
- Real-time (live) or offline (extract) analysis.
- Brings together data from different platforms.
- Simplifies data preparation.
- Extracts improve performance.

Example

Sales data is in an **Excel** file and customer details are in a **MySQL** database. Tableau connects to both, relates them using a common field (**Customer ID**) and builds dashboards of sales performance, customer trends and regional analysis.

6. Data Connectors

A **data connector** is the mechanism that lets Tableau connect to a data source and bring its data in. It acts as a **bridge** between Tableau and external data.

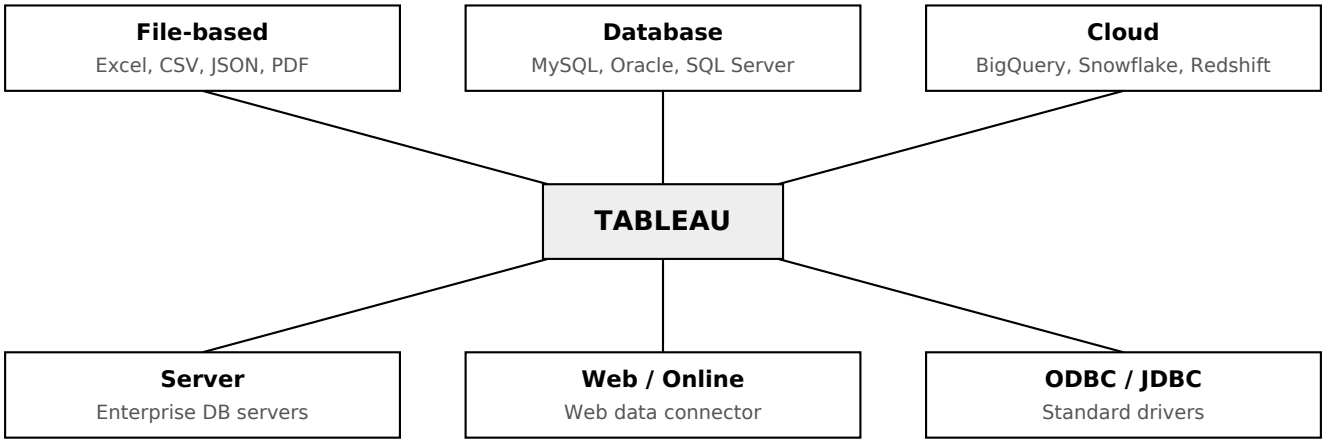


Figure: Types of data connectors (the connector is the bridge to Tableau)

Types of data connectors

1. **File-based connectors**: connect to files on a computer or network. Examples: Excel, CSV/Text, JSON, PDF, spatial files.
2. **Database connectors**: connect directly to relational databases. Examples: MySQL, SQL Server, Oracle, PostgreSQL, IBM DB2, SQLite.
3. **Cloud-based connectors**: connect to cloud platforms. Examples: Google BigQuery, Amazon Redshift, Snowflake, Azure SQL Database, Salesforce.
4. **Server connectors**: access data on enterprise database servers; used for centralized, large-scale data.
5. **Web and online connectors**: access data from online services and web platforms (e.g. Web Data Connector).
6. **ODBC and JDBC connectors**: **ODBC** (Open Database Connectivity) is a standard way to connect using an ODBC driver. **JDBC** (Java Database Connectivity) connects using Java-based drivers.

Common connectors

Connector	Type	Typical use
Excel	File	Spreadsheet analysis
CSV	File	Text / tabular data
MySQL	Database	Relational database analysis
SQL Server	Database	Enterprise data
Oracle	Database	Large organizational databases
PostgreSQL	Database	Open-source relational data
Salesforce	Cloud	CRM data
Snowflake	Cloud	Cloud data warehouse
Google BigQuery	Cloud	Large-scale analytics
Amazon Redshift	Cloud	Cloud data warehouse

How a connector works

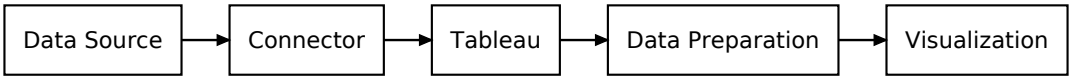
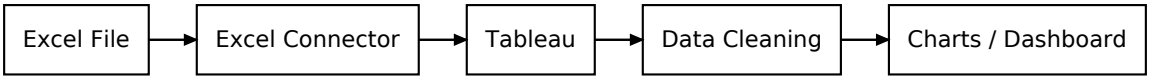


Figure: How a data connector works



Example

Connection modes

After connecting, you choose **Live** (Tableau queries the original source every time) or **Extract** (Tableau makes a copy in its extract format for faster, offline analysis).

7. Accessing Files

Files are the most common data sources in Tableau. Tableau can connect to many file formats **directly, without a database**.

File types supported

File type	Description
Excel (.xlsx)	Excel worksheets and tables
CSV (.csv)	Comma-separated text data
Text (.txt)	Tab-delimited or other text files
JSON (.json)	Structured JSON data
PDF (.pdf)	Tables extracted from PDF files
Microsoft Access (.mdb, .accdb)	Access databases
Spatial files	Shapefile, KML, GeoJSON for maps

Steps to access a file

1. Open Tableau Desktop.
2. On the Start page go to **Connect → To a File**.
3. Choose the file type (e.g. Microsoft Excel or Text File).
4. Browse to the file.
5. Select it and click **Open**.
6. The **Data Source page** opens.
7. Select the worksheet/table you need.
8. Drag the tables into the data model.
9. Check fields and data types.
10. Click **Sheet 1** to start building charts.

Example

File: Sales_Data.xlsx with columns Order ID | Product | Region | Sales | Profit.

Path: **Connect → To a File → Microsoft Excel → Sales_Data.xlsx**. Tableau shows the worksheets in the file; pick one and use Region, Sales and Profit to build charts.

File connection types

- **Live**: Tableau reads the original file whenever needed.
- **Extract**: Tableau creates a **.hyper** copy, which is faster and works offline.

Important points

- Keep the file accessible when using a live connection.
- If the file is moved or renamed, Tableau may lose the connection.
- To share a workbook together with its file data, save it as a **Packaged Workbook (.twbx)**.
- Always check data types, field names, missing values and duplicate records after connecting.

8. Connecting to a Text File

Tableau can connect directly to text-based files such as **CSV, TXT and TSV**.

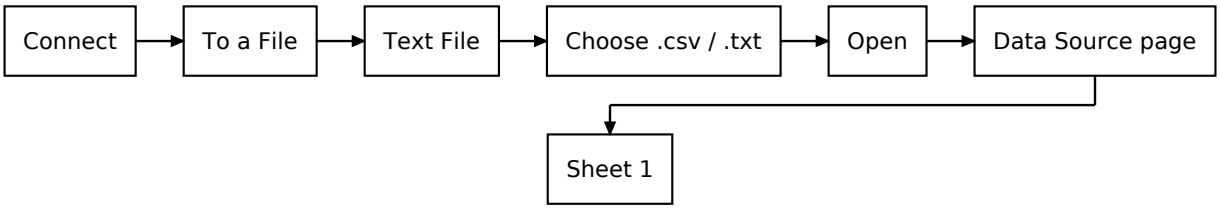


Figure: Connecting a text file

Steps

1. Open Tableau Desktop.
2. On the Start page, go to **Connect → To a File**.
3. Select **Text File**.
4. Browse to the .csv, .txt or .tsv file.
5. Select the file and click **Open**.
6. Tableau shows the **Data Source page**.
7. Check that Tableau correctly identified field names, data types, delimiters (the separator, such as a comma), rows and columns.
8. If needed, change data types or rename fields.
9. Go to a worksheet to start building charts.

Example: Sales.csv

Order ID	Product	Region	Sales
101	Laptop	North	50000
102	Mobile	South	30000
103	Tablet	East	25000
104	Laptop	West	45000

Possible charts: Sales by Region, Sales by Product, Total Sales, regional comparison, product-wise dashboard.

Important options on the Data Source page

- **Data Interpreter**: helps Tableau clean and read structured data in spreadsheet or text files.
- **Field / data types**: String, Number, Date, Date & Time, Boolean.
- **Filters**: filter data before using it.
- **Relationships / Joins**: combine the text file with other tables or files.

Advantages

- Simple and easy to use.
- No database server needed.
- Good for small and medium datasets.

- Easy to share and transfer.
- Useful for learning and basic analysis.

Limitations

- Large text files can be slow.
- Data must be properly structured.
- Changes in the file may need a refresh in Tableau.
- No advanced database features (security, indexing, etc.).

9. RData Files

An **RData file (.RData or .rda)** is a file format of the **R** programming language. It stores R objects such as data frames, variables and models.

Can Tableau open RData directly?

No. Tableau does not normally connect directly to .RData or .rda files. You first **export the data from R** into a format Tableau can read, such as **CSV or Excel**.

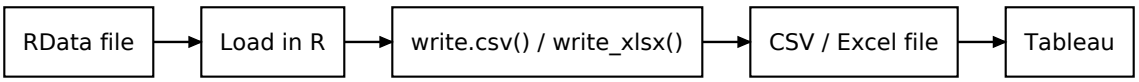


Figure: Getting RData into Tableau

Method 1: Export to CSV

```
load("sales_data.RData")
write.csv(sales_data, "sales_data.csv", row.names = FALSE)
```

Then in Tableau: **Connect → To a File → Text File → sales_data.csv**

Method 2: Export to Excel

```
install.packages("writexl")
library(writexl)
write_xlsx(sales_data, "sales_data.xlsx")
```

Then in Tableau: **Connect → To a File → Microsoft Excel → sales_data.xlsx**

Method 3: Connect Tableau with R (Rserve)

For advanced work, Tableau can talk to R through **Rserve**. This is used to run R's statistical calculations or predictive models from Tableau, not just to import a file.

RData vs Tableau

Feature	RData	Tableau
File extension	.RData, .rda	.twb, .twbx, .hyper
Main purpose	Store R objects	Data visualization and analytics
Direct Tableau connection	Generally not supported	n/a
Exchange format	CSV / Excel	CSV / Excel / Database / Hyper
Advanced integration	Rserve	Tableau + R

10. Connecting to MySQL

MySQL is a relational database management system where data is stored in databases, tables, rows and columns. Tableau can connect to MySQL directly.

Steps

1. Open Tableau Desktop.
2. On the Start page, under **Connect**, select **MySQL**.
3. Install the MySQL driver if Tableau asks for it.
4. Enter the connection details:
   - **Server**: server address, e.g. **localhost**
   - **Port**: usually **3306**
   - **Database**: name of the database
   - **Username** and **Password**
5. Click **Sign In**.
6. Tableau shows the available databases and tables.
7. Drag the required table to the Data Source area.
8. Add more tables if needed and create relationships, joins or unions.
9. Click **Sheet 1** to start building charts.

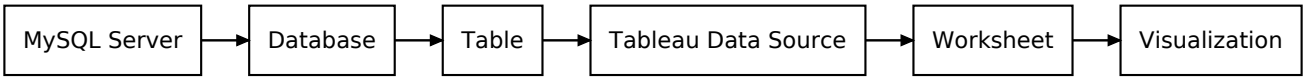


Figure: MySQL connection flow (e.g. localhost:3306 → SalesDB → Sales)

Example

Database **SalesDB** has a table **Sales**:

Order_ID	Product	Region	Sales	Profit
101	Laptop	North	50000	8000
102	Mobile	South	30000	5000
103	Tablet	East	20000	3000

Path: Connect → MySQL → Server → Database → Sales table → Sheet 1. Then build Sales by Region, Profit by Product, Monthly Sales.

Live vs Extract for MySQL

- **Live**: queries MySQL directly, shows database changes, needs access to the MySQL server, good when data must stay current.
- **Extract**: makes a Hyper copy, gives better performance, can be refreshed periodically, good for large data or when constant access is not needed.

11. Databases

A **database** is a structured collection of data stored in a **Database Management System (DBMS)**. Tableau connects to databases to build charts, dashboards and reports directly from an organization's data.

Databases Tableau can connect to

Database	Type
MySQL	Open-source relational database
Microsoft SQL Server	Enterprise relational database
Oracle Database	Enterprise database for large applications
PostgreSQL	Open-source relational database
Amazon Redshift	Cloud data warehouse
Snowflake	Cloud data platform
Google BigQuery	Google's cloud data warehouse
Azure SQL Database	Microsoft's cloud relational database
Teradata	Enterprise data warehouse
SAP HANA	In-memory database platform
MongoDB	NoSQL database (through connectors)

Connecting a database

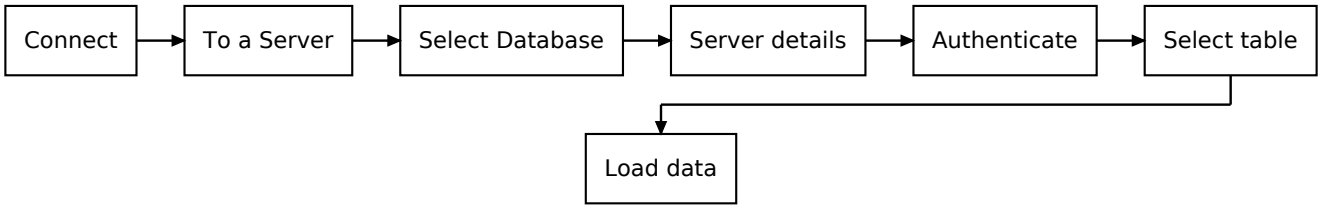


Figure: Connecting a database to Tableau

Details needed: server name or IP address, port number, database name, username, password, authentication method.

Live vs Extract

Mode	Advantages	Disadvantages
Live	Up-to-date data; no full copy stored; good for changing data	Speed depends on database and network; complex queries can be slow
Extract	Faster; less dependence on the database; good for heavy analysis; can work offline	Not current until refreshed; needs storage space

What you can do with database tables

Select tables, relate them, join them, union them, write custom SQL, apply filters and create calculated fields.

Relationship example

Customer_ID	Customer_Name	City
101	Ravi	Delhi

Customer_ID	Customer_Name	City
102	Anil	Mumbai

Customer_ID	Product	Sales
101	Laptop	50000
102	Printer	20000

The Customer table (top) and Sales table (bottom) are related on **Customer_ID**, so you can analyse sales by customer or city.

Custom SQL

You can write your own SQL query to fetch exactly the data you need:

```
SELECT Customer_Name, City, Sales
FROM Customer
JOIN Sales
ON Customer.Customer_ID = Sales.Customer_ID;
```

Database security

Username/password, integrated authentication, database permissions, encryption, row-level security and user access controls.

Database vs Tableau data source

Database	Tableau data source
Stores organizational data	Defines how Tableau accesses and uses the data
Managed by a DBMS	Managed through Tableau
Made for storage and transactions	Made for analysis and visualization
Can contain many tables	Contains only selected tables and relationships
e.g. MySQL, Oracle, SQL Server	e.g. .tds, .tdsx, or a connection in a workbook

> ***Remember:*** *Databases let Tableau work directly on company data without copying it into spreadsheets.*

12. NoSQL Databases

NoSQL (Not Only SQL) databases store and process large volumes of **unstructured, semi-structured or rapidly changing** data. They do not need fixed tables. Tableau can work with NoSQL data, but it depends on the available connector or driver.

Common NoSQL databases

NoSQL database	Data model	Tableau connectivity
MongoDB	Document	Supported connectors / drivers / partners
Cassandra	Wide-column	Connectors / drivers
Amazon DynamoDB	Key-value / document	Cloud or partner connectivity
Couchbase	Document / key-value	Supported connectivity
HBase	Wide-column	ODBC-based solutions

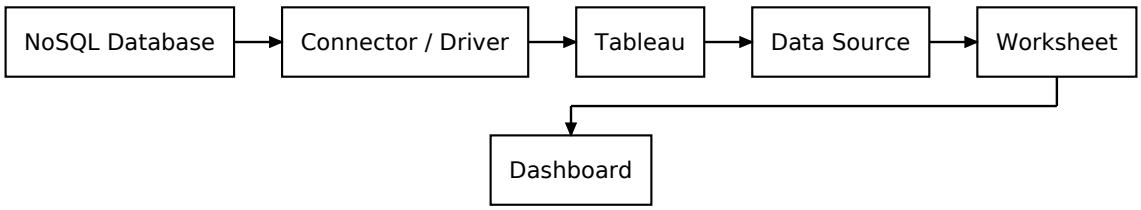


Figure: NoSQL to Tableau

Steps (e.g. MongoDB)

1. Open Tableau Desktop and go to **Connect**.
2. Select the available connector.
3. Enter database/server details and login information.
4. Select the required **collection** (the NoSQL word for a table).
5. Review data on the Data Source page and handle nested fields.
6. Create worksheets and dashboards.

Example MongoDB document

```
{
  "customer": "Rahul",
  "city": "Bangalore",
  "product": "Laptop",
  "sales": 75000
}
```

Possible charts: sales by city, sales by product, customer-wise sales, sales trends.

Why use NoSQL with Tableau?

- Handles very large volumes of data.
- Works with semi-structured and unstructured data.
- Supports changing data structures.
- Useful for Big Data analytics.
- Can be combined with other data sources in Tableau.

SQL vs NoSQL

SQL databases	NoSQL databases
Data in structured tables (rows and columns)	Data as documents, key-value pairs, graphs or wide columns
Fixed, predefined schema	Flexible schema
e.g. MySQL, Oracle	e.g. MongoDB, Cassandra

Part C: Live Connection, Extracts and In-Memory

13. Live Connection

A **live connection** means Tableau connects **directly** to the original data source and gets data from it in real time, whenever a chart or dashboard needs it. Tableau does **not** import the whole data.

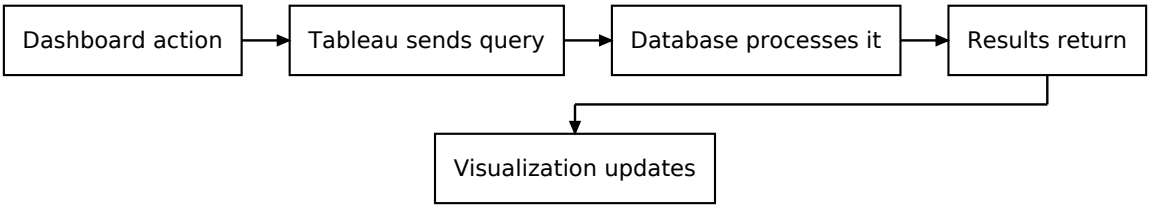


Figure: How a live connection works

When you apply a filter or change a view, Tableau sends a **query** to the database. The database processes it and returns the results.

Key features

1. **Real-time data**: shows the latest data.
2. **No full import**: data stays in the original database.
3. **Database processing**: the database server does the work.
4. **Good for large data** that is too big to load into Tableau.
5. **Automatic updates**: database changes appear without refreshing an extract.
6. **Needs the database to be available**.
7. **Speed depends on the database** and network.

Example

A company's sales database is updated all the time. A live dashboard shows the latest sales transactions, inventory, customer orders, revenue and regional performance.

Advantages

- Near-real-time data.
- No duplicate copy of data.
- Good for frequently changing data.
- Can handle very large databases.
- Data management stays centralized in the database.

Disadvantages

- Performance depends on the database and network.
- Database must be available.
- Complex dashboards send many queries.
- Increases workload on the database server.
- Offline analysis is not possible.

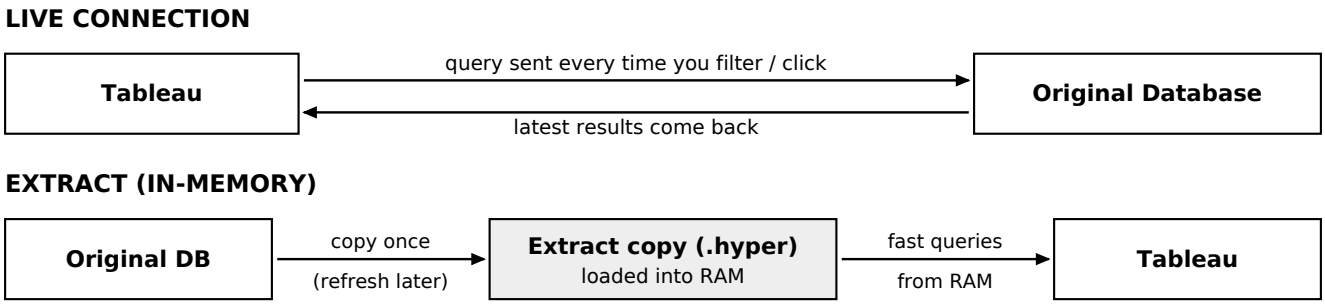


Figure: Live connection vs Extract

Live connection vs Extract

Feature	Live connection	Extract
Data location	Original database	Tableau extract (.hyper)
Freshness	Near real-time	As of last refresh
Performance	Depends on database	Often faster
Network dependency	Higher	Lower after extract is made
Database load	Higher	Lower
Offline access	Limited	Better
Refresh needed	No	Yes

14. Data Extraction

Data extraction is creating a **copy of the data** from the original source and storing it in Tableau's optimized **Hyper extract format (.hyper)**. Tableau then works on this copy instead of asking the original source every time.

Why use extraction?

- Better performance with large datasets.
- Faster filtering, aggregation and charts.
- Analysis possible even when the original source is unavailable.
- Fewer queries to the database.
- Data can be filtered before creating the extract.
- Extract can be refreshed manually or on a schedule.

Steps to create an extract

1. Connect to a data source (Excel, CSV, MySQL...).
2. Go to the **Data Source page**.
3. Select **Extract** instead of **Live**.
4. Click Extract / Edit to set extract settings.
5. Optionally add filters to reduce data.
6. Tableau creates the extract (a **.hyper** file).
7. Open a worksheet and build charts.

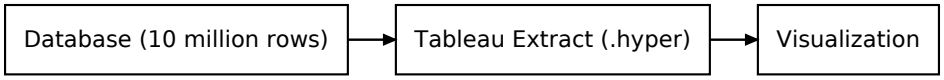


Figure: Data extraction

Example

A database has 10 million sales records. Instead of querying all of them again and again, Tableau makes an extract and analyses that, which is usually much faster.

Types of extract refresh

- **Full refresh**: the whole extract is replaced with fresh data.
- **Incremental refresh**: only new records or changes are added, so less data is processed.

Extract filter example

Database has sales from 2015–2026 but you only need 2024–2026. Add an extract filter **Year = 2024, 2025, 2026**. The extract becomes smaller and faster.

Extract file

Modern extracts use the Hyper format, e.g. **Sales_Data.hyper**. The Hyper engine is built for analytical queries on large data.

Live vs Extract

Live connection	Extract
Queries the original data source	Uses a copy of the data
Data can be very current	Current only after refresh
Depends on source availability	Works without continuous source access
May be slower for complex analysis	Often faster
No extract file	Creates a .hyper file

Remember: *Extract = an optimized copy of the data in a .hyper file.*

15. In-Memory

In-memory means storing data in the computer's **RAM** (Random Access Memory) so Tableau can read and analyse it very quickly. Tableau uses its **Hyper engine** for in-memory processing when working with extracts.

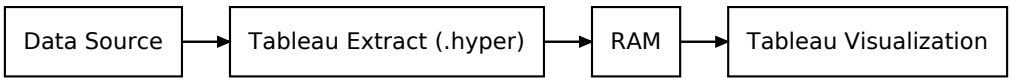


Figure: How in-memory works

When data is imported as an extract, it is stored in an optimized Hyper file. During analysis Tableau loads the needed data into memory and answers queries fast.

Key features

1. **Fast processing**: much faster than fetching from a remote database each time.
2. **High-performance analytics**: quick calculations, filtering, aggregation and sorting.
3. **Columnar storage**: Hyper stores data column by column, which suits analysis.
4. **Less database dependency**: no need to query the original database for every chart.
5. **Supports large datasets**.
6. **Extract refresh**: the extract can be refreshed to get new data.

Example

10 million sales records in a SQL database: SQL Database → Tableau Hyper Extract → Tableau → Dashboard. Tableau analyses the extract without sending every query to SQL.

Advantages

- Faster dashboards.
- Efficient filtering and aggregation.
- Fewer repeated queries to the source.
- Good for interactive dashboards.
- Helps when the original database is slow or far away.

Limitations

- Not real-time unless refreshed often.
- Extracts need storage space.
- Large extracts need a lot of memory and processing power.
- Changes in the source are not shown until refresh.

Live vs In-memory extract

Feature	Live connection	In-memory / Extract
Data source	Original database	Tableau Hyper extract
Freshness	Near real-time	Depends on refresh
Query processing	Mainly the database	Tableau Hyper engine
Dashboard speed	Depends on database / network	Often faster
Database dependency	High	Lower
Offline analysis	Limited	More suitable
Best for	Frequently changing data	Fast interactive analysis

Part D: Organizing and Combining Data

16. Data Layer

The **data layer** is the underlying structure of data that Tableau uses to **connect, combine, prepare and analyse** information. It is the **foundation** of every visualization and dashboard.

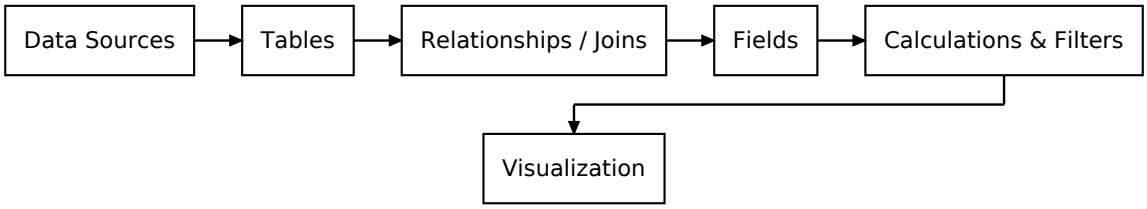


Figure: Data layer flow

Components of the data layer

1. **Data sources**: databases, spreadsheets, cloud services, files (Excel, CSV, MySQL, SQL Server, Oracle, Google Sheets).
2. **Tables**: the actual data in rows and columns (customers, sales, products, employees...).
3. **Relationships**: connect tables on common fields (e.g. Customer ID links Customer and Orders tables).
4. **Joins**: combine tables on a common field. Types: Inner, Left, Right, Full Outer.
5. **Unions**: combine rows of tables with the same structure (e.g. January Sales + February Sales).
6. **Dimensions and measures**: dimensions describe/group (Product, Region, Customer); measures are numbers to aggregate (Sales, Profit, Quantity).
7. **Data types**: text, number, date, Boolean, etc. Correct types give correct analysis.
8. **Calculated fields**: new fields from formulas, e.g. **Profit Ratio = SUM([Profit]) / SUM([Sales])**.
9. **Filters**: restrict data, e.g. only 2026 or only one region.

Example

Customer ID	Product	Region	Sales	Profit
101	Laptop	North	50,000	8,000
102	Mobile	South	30,000	5,000
103	Laptop	West	45,000	7,000

From this, Tableau can create Sales by Region, Profit by Product, Monthly Sales Trend, Top Customers and Profit Margin.

17. Tables

A **table** is data arranged in **rows and columns**. Tableau connects to tables from databases, spreadsheets and other sources.

Types of tables

- **Database tables**: stored in MySQL, Oracle, SQL Server, PostgreSQL.
- **Spreadsheet tables**: each Excel worksheet is usually one table.
- **Logical tables**: in the relationships model, tables are connected logically on common fields without merging all rows immediately.
- **Physical tables**: used inside a logical table to create joins and unions.

Structure of a table

- **Rows**: individual records.
- **Columns**: fields/attributes.
- **Field names**: column names.
- **Data values**: the actual information in the cells.

Student_ID	Name	Department	Marks
101	Amit	CSE	85
102	Ravi	AI	90
103	Priya	DS	88

Student_ID, Name, Department and Marks are columns; each student is a row.

Ways to combine tables

- **Relationships**: connect tables on related fields, keeping each table's structure.
- **Joins**: add columns from two or more tables using a common field.
- **Unions**: add rows from tables with similar columns.

JOIN = adds COLUMNS (tables side by side, matched on ID)

Customer

ID	Name
101	Rahul
102	Priya

Sales

ID	Sales
101	50,000
102	30,000

→

Result (wider)

ID	Name	Sales
101	Rahul	50,000
102	Priya	30,000

UNION = adds ROWS (similar tables one below the other)

Jan Sales

Month	Sales
Jan	120

Feb Sales

Month	Sales
Feb	150

→

Result (longer)

Month	Sales
Jan	120
Feb	150

Figure: Join vs Union

Join types

Inner (only matching rows), Left (all left + matching right), Right (all right + matching left), Full Outer (all rows from both).

On the Data Source page you can

Select tables, create relationships, joins and unions, rename fields, change data types, filter data, use custom SQL and preview data.

Tables become visualizations

Text tables, crosstabs, bar charts, line charts, pie charts, heat maps, highlight tables and dashboards.

Text table

A text table shows exact values in rows and columns like a spreadsheet:

Region	Product	Sales
North	Laptop	₹50,000
South	Laptop	₹65,000
North	Printer	₹25,000
South	Printer	₹30,000

Use a text table when exact numbers matter more than a picture.

18. Dimensions

Dimensions are fields with **qualitative or categorical** information that **describe and classify** data. They are used to group, segment and organize data. Examples: Customer Name, Product, City, Country, Department, Category, Date.

Examples

Dimension	Example values
Customer Name	Amit, Ravi, Priya
Product	Laptop, Printer, Mobile
Category	Technology, Furniture
City	Delhi, Mumbai, Bengaluru
Country	India, USA, UK
Department	CSE, AI, DS
Region	North, South, East, West

In the Data Pane

Dimensions are listed in the Dimensions section (above the Measures). They are usually shown as **blue** fields because they are **discrete**.

Purpose of dimensions

Categorize data, group records, filter information, create headers, define the level of detail, compare categories, organize the view.

Example

Product	Region	Sales
Laptop	North	50,000
Laptop	South	60,000
Printer	North	25,000
Printer	South	30,000

Product and Region are **dimensions**; Sales is a **measure**. Putting Region in the view groups sales into North and South.

Discrete dimensions

A discrete field makes separate categories, e.g. Region: North | South | East | West. Discrete fields are shown in **blue**.

Dimensions as filters

Put Department on Filters and choose CSE, AI, Data Science or Cyber Security; only those departments are shown.

Level of detail

Adding dimensions makes the view more detailed. **Sales by Region** gives one total per region. Adding Product gives **Sales by Region → Product**, which is more detailed.

Dimension vs Measure

Dimension	Measure
Describes data	Quantifies (counts/measures) data
Usually categorical	Usually numerical
Used for grouping and filtering	Used for calculation and aggregation
e.g. Product, Region, City	e.g. Sales, Profit, Quantity
Usually blue (discrete)	Usually green (continuous)

> ***Remember:*** *Dimension = Describes (who/what/where). Measure = Maths (how much/how many).*

19. Facts

Facts are data values that record **measurable business events** or numerical observations. They are usually stored in the **fact table** of a data warehouse and are analysed in Tableau as **measures**. Example: each sales transaction is a fact with Sales Amount, Quantity, Discount and Cost.

Examples

Fact	Example value
Sales Amount	₹50,000
Quantity Sold	25
Profit	₹8,000
Discount	10%
Cost	₹42,000
Order Amount	₹15,000

Fact table example

Order_ID	Product_ID	Customer_ID	Quantity	Sales	Profit
1001	P01	C101	2	50,000	8,000
1002	P02	C102	5	20,000	4,000
1003	P01	C103	1	25,000	5,000

Order_ID, Product_ID and Customer_ID are **keys/identifiers**. Quantity, Sales and Profit are **facts**.

Facts become Tableau measures

Numeric fact fields are automatically treated as measures and aggregated: Sales → SUM(Sales), Quantity → SUM(Quantity), Profit → SUM(Profit), Discount → AVG(Discount). Aggregation functions: **SUM, AVG, MIN, MAX, COUNT, MEDIAN**.

Fact table and dimension table

A data warehouse has **fact tables** (the measurements) and **dimension tables** (descriptions of those measurements), linked by keys such as Product_ID.

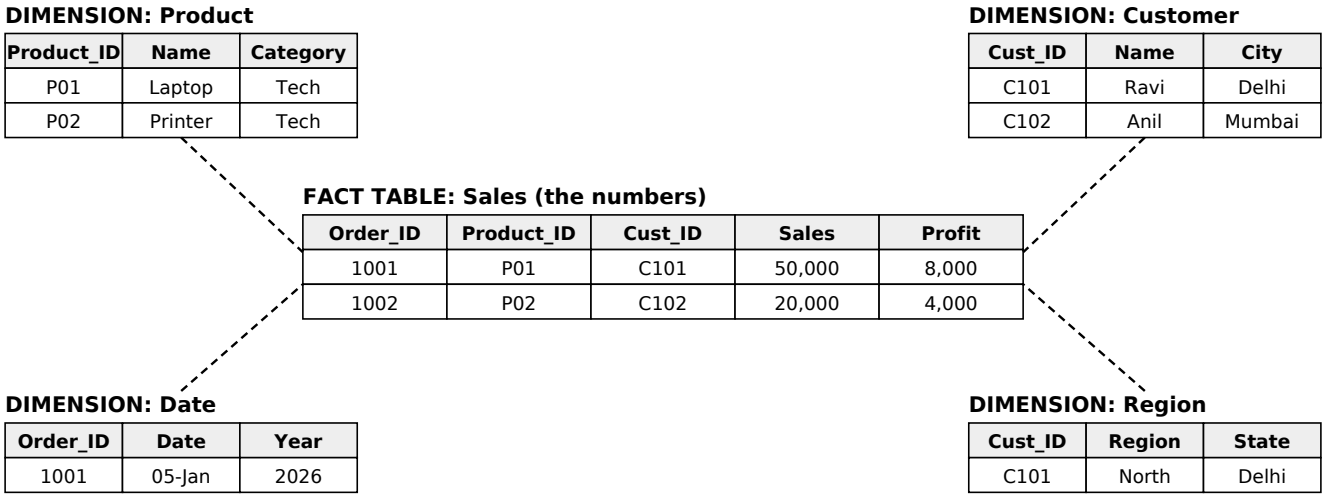


Figure: Star schema. The fact table in the centre holds the numbers (facts); the dimension tables around it hold descriptions and link to it through ID keys.

Product_ID	Product_Name	Category
P01	Laptop	Technology
P02	Printer	Technology

Above: a Product dimension table that describes the Product_IDs used in the fact table.

Facts vs Dimensions

Facts	Dimensions
Measurable values	Descriptive information
Usually numerical	Usually categorical
Used for calculation and aggregation	Used for grouping and filtering
e.g. Sales, Profit, Quantity	e.g. Product, Region, Customer
Stored in fact tables	Stored in dimension tables
Usually become Tableau measures	Usually become Tableau dimensions

Importance

Facts give the **numbers** for analysis: KPIs, charts, performance comparison and trends. Region + Sales → Sales by Region; Product + Profit → Profit by Product; Year + Sales → Sales Trend.

20. Joins

A **join** combines data from two or more tables based on a **common field**. It is used when related information is stored in different tables.

Example tables (common field = Customer_ID)

Customer_ID	Customer_Name	City
101	Rahul	Delhi
102	Priya	Mumbai
103	Amit	Bangalore

Customer_ID	Product	Sales
101	Laptop	50000
102	Mobile	30000
104	Tablet	20000

First table = **Customer** (left), second = **Sales** (right). Join condition: Customer.Customer_ID = Sales.Customer_ID.

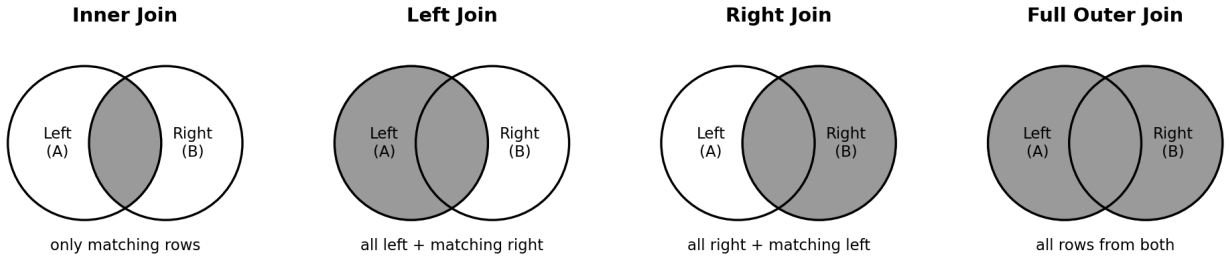


Figure: The four join types (shaded part = rows kept in the result)

1. Inner join: only matching records from both tables

Customer_ID	Customer_Name	Product	Sales
101	Rahul	Laptop	50000
102	Priya	Mobile	30000

103 and 104 are left out because they have no match. Use when you need only matching data.

2. Left join: all rows of the left table + matching rows of the right

Customer_ID	Customer_Name	Product	Sales
101	Rahul	Laptop	50000
102	Priya	Mobile	30000
103	Amit	NULL	NULL

Use when every record of the main (left) table must be kept.

3. Right join: all rows of the right table + matching rows of the left

Customer_ID	Customer_Name	Product	Sales
101	Rahul	Laptop	50000
102	Priya	Mobile	30000
104	NULL	Tablet	20000

Use when every record of the right table must be kept.

4. Full outer join: all rows from both tables (NULL where no match)

Customer_ID	Customer_Name	Product	Sales
101	Rahul	Laptop	50000
102	Priya	Mobile	30000
103	Amit	NULL	NULL
104	NULL	Tablet	20000

Use when no record from either table should be lost.

Steps to create a join

1. Connect to the data source.
2. Open the Data Source page.
3. Drag the first table onto the canvas.
4. Drag the second table next to it (open the table to reach the join area).
5. Choose the join type: Inner, Left, Right or Full Outer.
6. Select the common field.
7. Add more join conditions if needed.
8. Start building the visualization.

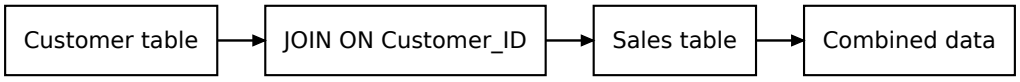


Figure: Joining two tables

Join vs Relationship

Join	Relationship
Combines tables physically at the data-source level	Defines a logical connection between tables
You choose a join type	Tableau decides how to query the tables
Can create duplicate rows (many-to-many)	Keeps each table's level of detail better
Good for detailed row-level combining	Preferred in Tableau's logical data model

21. Data Blending

Data blending combines data from **two or more different data sources** in a single visualization. It is useful when the sources **cannot be joined directly**. Example: sales in an Excel file and targets in a MySQL database, blended on the common field **Region**.

How it works

Blending uses a **primary data source** and one or more **secondary data sources**.

- **Primary source**: the data source used first in the view; gives the main dimensions and measures.
- **Secondary source**: the extra source whose data is blended in; usually adds measures.

Tableau links them on a common field and combines the **aggregated** results.

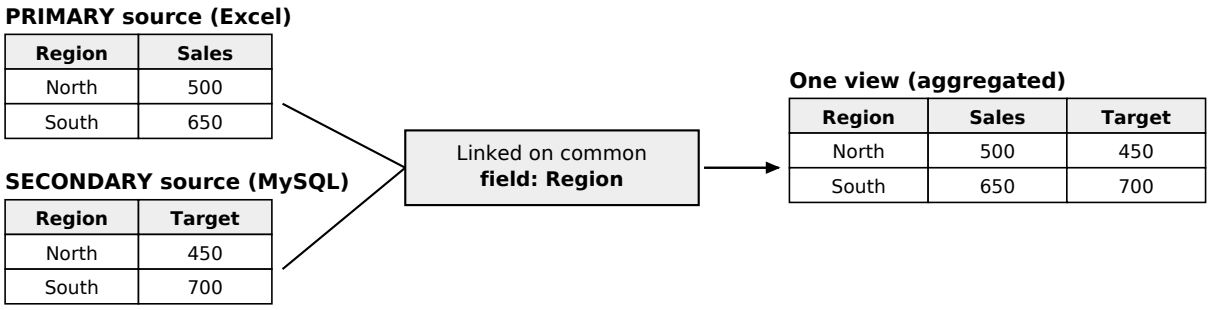


Figure: Data blending, with a primary and a secondary source shown together

Steps

1. Connect to the first data source.
2. Connect to the second data source.
3. Build a worksheet using the primary source.
4. Add a field from the secondary source.
5. Tableau finds the common (linking) field.
6. The common field sets up the blend relationship.
7. Tableau shows aggregated results from both sources in the view.

Example

Primary: Region → Sales. Secondary: Region → Target. After blending: **Region → Sales → Target**, which lets you compare actual sales with targets.

Advantages

- Combines data from different sources.
- Useful when joins are not possible.
- Works with databases, Excel files and more.
- Compares metrics from different systems.
- Does not physically merge the underlying data.

Data blending vs Joins

Data blending	Joins
Combines data from different data sources	Usually combines tables within one data source
Uses primary and secondary sources	Uses tables with join conditions
Combines at the visualization / aggregated level	Combines at the row level
Useful when sources cannot be joined directly	Useful when tables have compatible structures
Uses common linking fields	Uses join keys

> ***Remember:*** *Blending combines summarized data from different sources using a common field such as Region, Product, Customer ID or Date.*

Part E: Tableau Files

22. Tableau Workbook

A **Tableau workbook (.twb or .twbx)** is the main file of Tableau Desktop, used to create, save and share visualizations and dashboards.

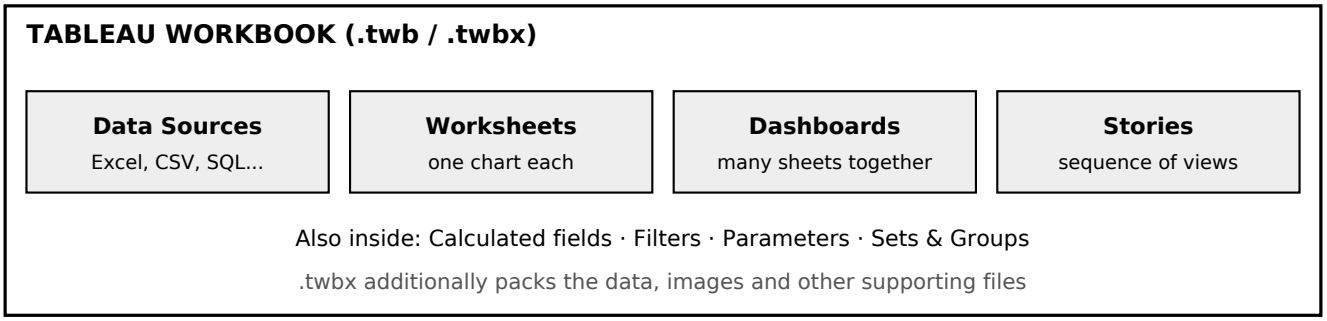


Figure: What a Tableau workbook contains

Components of a workbook

1. **Data sources**: connections to Excel, CSV, SQL databases, cloud services, etc.
2. **Worksheets**: individual charts, graphs or tables.
3. **Dashboards**: many worksheets combined into one interactive view.
4. **Stories**: a sequence of dashboards/worksheets that tells a data story.
5. **Calculated fields**: custom formulas.
6. **Filters**: restrict the data shown.
7. **Parameters**: user-controlled variables for interactivity.
8. **Sets and groups**: organize and categorize data.

Workbook file types

File type	Extension	Description
Tableau Workbook	.twb	XML file with the workbook structure only; does NOT contain the data
Tableau Packaged Workbook	.twbx	Workbook + data sources + images + supporting files in one package; easy to share

Advantages

- Many visualizations in one file.
- Supports interactive dashboards and stories.
- Easy to share as .twbx.
- Analysis with filters, parameters and calculated fields.
- Can connect to many data sources.

Typical workflow

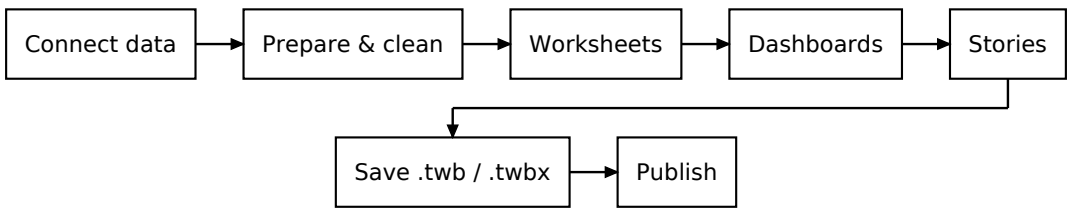


Figure: Typical workbook workflow

1. Connect to data. 2. Prepare/clean. 3. Create worksheets. 4. Build dashboards. 5. Create stories (optional). 6. Save as .twb or .twbx. 7. Publish to Tableau Server or Tableau Cloud.

23. Tableau Packaged Data Source

A **Tableau Packaged Data Source (.tdsx)** is a data source file that contains the **data source definition + a local copy of the data (extract) + supporting files**. It can be shared without needing access to the original database or file.

Key features

- Connection and extract are stored in one file.
- Can be shared with other Tableau users.
- Allows offline access.
- Keeps calculated fields, groups, sets, hierarchies and metadata.
- Makes collaboration easy.

File types

File type	Extension	Description
Tableau Data Source	.tds	Only the data source definition; no data
Tableau Packaged Data Source	.tdsx	Definition + data extract + supporting files

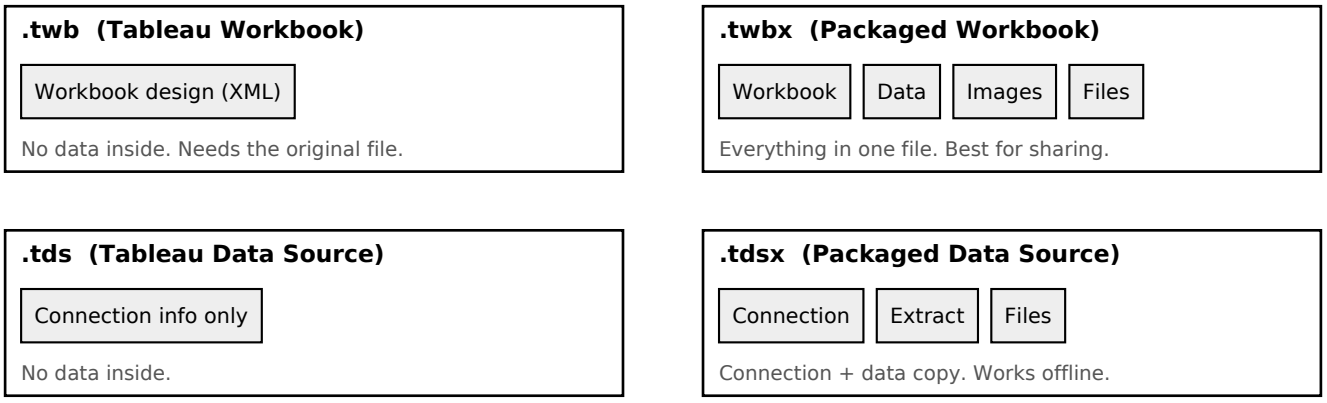


Figure: Plain files vs packaged files (the x means everything is packed inside)

Components

Connection information, data extract, calculated fields, groups and sets, hierarchies, metadata (field names, aliases, data types), supporting files.

Advantages

- Easy to share with colleagues.
- No direct access to the original database needed.
- Offline analysis.
- Everyone uses the same version of the data.
- Less dependence on external sources.

Limitations

- Data becomes outdated unless the extract is refreshed.
- File can be large for big datasets.
- Changes in the original source are not reflected automatically.

Example

An analyst connects to an Excel sales file, creates an extract and saves it as a **.tdsx**. Team members open it in Tableau Desktop without needing the original Excel file.

All Tableau file types at a glance

Extension	Name	Contains
.twb	Tableau Workbook	Sheets, dashboards, stories (XML); no data
.twbx	Packaged Workbook	Workbook + data + images; for sharing
.tds	Tableau Data Source	Connection information only
.tdsx	Packaged Data Source	Connection + extract + files
.hyper	Extract file	The extracted data (Hyper engine format)

Remember: *The "x" at the end means everything is packed inside.*

Part F: The Tableau Workspace

24. Tableau Workspace

The **Tableau workspace** is the main working screen where you connect to data, build charts, analyse information and make dashboards and stories.

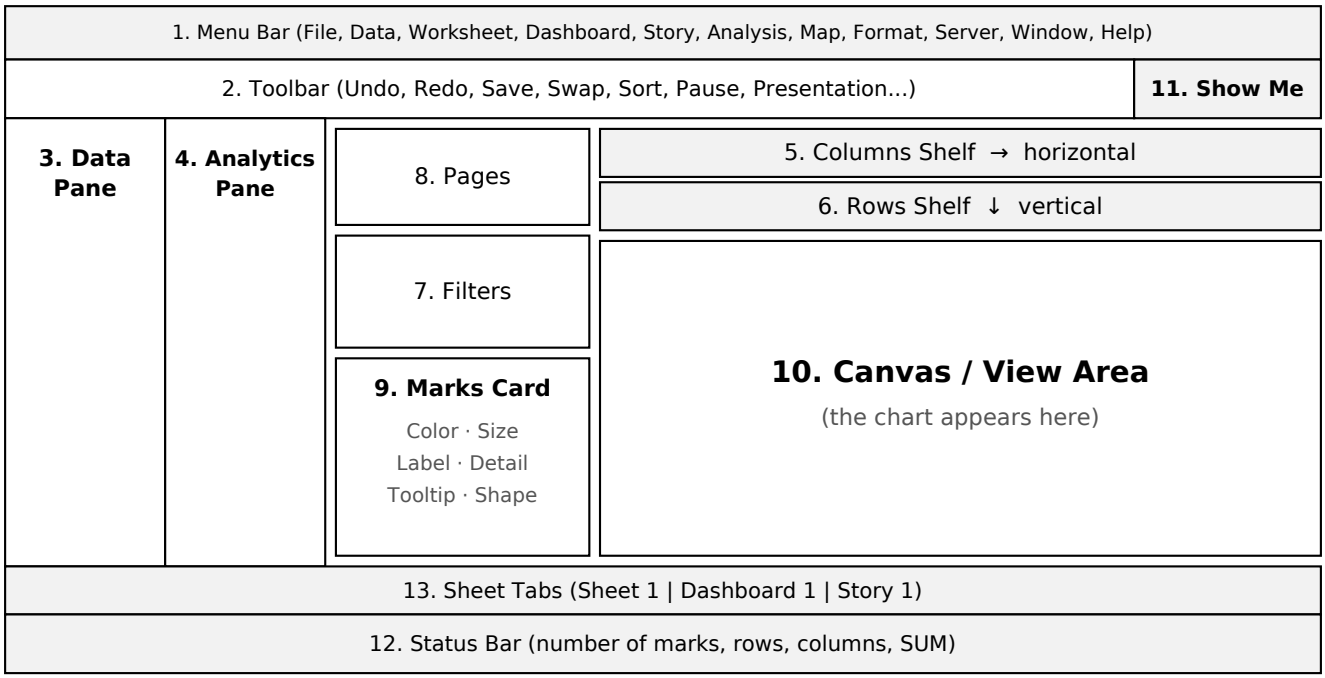


Figure: Layout of the Tableau workspace (numbers match the list)

Components

No.	Component	What it does
1	Menu Bar	Commands for files, data, worksheets, dashboards, analysis, formatting and help
2	Toolbar	Frequently used tools: Undo, Redo, Save, Add Data Source, Sort, Filter, Presentation
3	Data Pane	Shows dimensions, measures, parameters and sets of the data source
4	Analytics Pane	Analytics objects: average line, median with quartiles, trend line, forecast, reference line, clusters
5	Columns Shelf	Horizontal arrangement of fields
6	Rows Shelf	Vertical arrangement of fields
7	Filters Shelf	Restricts the data shown
8	Pages Shelf	Splits the view into a sequence of pages
9	Marks Card	Controls how data looks: Color, Size, Label, Detail, Shape, Tooltip
10	Canvas / View	Central area where the chart is built and shown
11	Show Me	Suggests chart types for the selected fields
12	Status Bar	Shows number of marks, selected data and summary statistics
13	Sheet Tabs	At the bottom; switch between worksheets, dashboards and stories

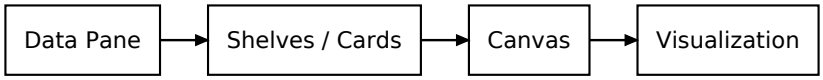


Figure: Basic structure of making a chart

Example: drag **Sales** to Columns and **Category** to Rows, and Tableau draws a bar chart automatically.

Remember: *The workspace is the complete environment for data analysis, visualizations, dashboards and stories.*

25. Menus

The **Menu Bar** at the top of Tableau has commands for creating, managing, analysing, formatting and publishing visualizations. Tableau Desktop has **11 menus**.

Menu	Purpose	Important options
1. File	Manage workbooks and files	New, Open, Save, Save As, Close, Export, Print
2. Data	Manage data sources	Connect to Data, New Data Source, Refresh, Edit Relationships, Extracts, Replace data source
3. Worksheet	Manage worksheets	New Worksheet, Duplicate, Clear, Show/Hide Cards, Show Summary, Export
4. Dashboard	Create dashboards	New Dashboard, Add Sheets, Actions, Layout / device layouts, Export
5. Story	Present data as a story	New Story, Add Story Point, Duplicate, Format, Export
6. Analysis	Perform analysis	Aggregate Measures, % of Total, Totals, Trend Lines, Forecast, Reference Lines, Calculated Field, Parameter, Table Calculations
7. Map	Geographic data	Map Layers, Background Maps, Geocoding, Map Options, Routes
8. Format	Appearance	Font, Alignment, Borders, Shading, Lines, Cell Size
9. Server	Publish and manage content	Sign In, Publish Workbook, Publish Data Source, Permissions, Refresh published data
10. Window	Manage the workspace	Show Toolbar, Show Status Bar, Switch between workbooks
11. Help	Assistance	Tableau Help, Documentation, Tutorials, Keyboard Shortcuts, About Tableau

Key points

- A **worksheet** has one chart; a **dashboard** combines many worksheets; a **story** is a sequence of worksheets/dashboards.
- The **Analysis** menu is used for deeper analysis (forecast, trend lines, calculations).
- The **Map** menu shows data like sales by state or customers by city on a map.
- The **Server** menu is used with Tableau Server / Tableau Cloud.
- Examples: **File → New** creates a workbook; **Window → Show Toolbar** shows or hides the toolbar.

Remember: *Order of menus: File, Data, Worksheet, Dashboard, Story, Analysis, Map, Format, Server, Window, Help ("Friendly Dogs Wag, Dance, Sing; Always Make Funny Sounds When Happy").*

26. Toolbar

The **toolbar** is the row of icons **below the Menu Bar**. It gives one-click access to frequently used commands.

Button	Function
New Workbook / Open / Save	Create, open or save a workbook
Undo	Reverses the last action
Redo	Re-applies an action that was undone
Reconnect	Reconnects to the data source
Pause Updates	Temporarily stops automatic updating of the view
Run Update	Manually updates the view after pausing
Swap	Exchanges the fields on Rows and Columns
Sort Ascending	Low to high / A to Z
Sort Descending	High to low / Z to A
Group	Combines selected marks into a group
Keep Only	Shows only the selected marks
Exclude	Removes the selected marks
Highlight	Highlights selected data, others stay visible
Show Me	Recommends chart types for the selected fields
Presentation Mode	Clean full-screen view for presenting

Important buttons explained

- **Save**: saves as **.twb** (workbook) or **.twbx** (packaged workbook).
- **Pause Updates / Run Update**: with large data, pause updates while making several changes, then click Run Update once. Improves performance.
- **Swap**: before, Rows = Sales, Columns = Region; after Swap, Rows = Region, Columns = Sales. Changes chart orientation quickly.
- **Group**: Bangalore + Chennai + Hyderabad → "South India".
- **Keep Only**: select Technology → only Technology data remains.
- **Exclude**: select Furniture → Furniture is removed.
- **Show Me**: suggests bar, line, pie, scatter plot, maps, treemap, heat map, box plot, histogram, Gantt chart.
- **Presentation Mode**: for classes, project demos, meetings.

Menu Bar vs Toolbar

Menu Bar	Toolbar
Contains many / all commands	Contains frequently used commands
Organized into text menus	Uses buttons / icons
e.g. Analysis → Forecast	e.g. Undo button
Detailed functionality	Quick access
Command-oriented	Action-oriented

27. Data Pane

The **Data Pane** is on the **left side** of the workspace. It shows all connected data sources, tables, fields and metadata. It is where data is **accessed and organized** before making charts.

Features

Feature	Meaning	Example
1. Data Source	Shows connected sources; switch between them; live or extract	Superstore file
2. Tables	Lists tables; expand to see fields	Orders, Returns
3. Dimensions	Qualitative / categorical fields, blue icon	Customer Name, Category, City, Region, Country
4. Measures	Quantitative / numeric fields, green icon	Sales, Profit, Quantity, Discount, Revenue
5. Hierarchies	Related dimensions in a drill-down order	Country → State → City
6. Sets	Custom subsets to compare with the rest; fixed or dynamic	Top 10 customers
7. Groups	Combine several values into one category	"North India" = Delhi, Punjab, Haryana, Himachal Pradesh
8. Parameters	User-defined variables (number, string, date, Boolean) for interactivity	Top-N selector
9. Calculated Fields	New fields from formulas	Profit Ratio = SUM([Profit]) / SUM([Sales])
10. Geographic Roles	Auto-detects location fields for maps	Country, State, Postal Code, Latitude, Longitude
11. Default Properties	Default number/date format, aggregation, colour, aliases, comments	Sales shown as ₹
12. Search Box	Quickly finds a field in big datasets	Type "prof"
13. Sort Options	Sort fields A–Z or by data source order	
14. Metadata	Data type, default aggregation, geographic role, field name, source	

Data types and icons

Data type	Icon	Example
String	Abc	Customer Name
Number (whole)	#	Quantity
Number (decimal)	#	Sales
Date	Calendar	Order Date
Date & Time	Calendar + clock	Transaction Time
Boolean	T/F	Is Returned
Geographic	Globe	Country

Common operations

Rename, hide/unhide, create calculated fields, parameters, sets, groups and hierarchies, change data types, assign geographic roles, change default properties, duplicate fields, split/custom split string fields, create bins.

Group vs Set

Group	Set
Merges values into one new category	Subset where members are IN or OUT
Simplifies categories	Compares selected data with the rest
Fixed	Fixed or dynamic

Advantages

- All fields organized in one place.
- Easy data exploration.
- Quick drag-and-drop.
- Calculated fields and parameters.
- Better data management.
- Interactive dashboards via sets, groups and parameters.
- Search and hierarchies make big data easy to navigate.

28. Analytics Pane

The **Analytics Pane** is on the left side, **next to the Data Pane** (Analytics tab). It has built-in analysis tools that you **drag and drop** onto a chart, with no formulas needed.

Features

Feature	Meaning
1. Constant Line	Fixed line at a value you enter, e.g. Sales Target ₹1,00,000, Pass Mark 40, Budget ₹5,00,000
2. Reference Line	Line based on data: Average, Median, Minimum, Maximum, Sum or custom
3. Distribution Band	Shaded range with upper and lower limits; shows variation/spread
4. Box Plot	Shows Minimum, Q1, Median, Q3, Maximum; highlights outliers
5. Trend Line	Shows relationship/direction and helps predict. Types: Linear, Logarithmic, Exponential, Polynomial, Power
6. Forecast	Predicts future values from past data (sales, revenue, demand, inventory)
7. Cluster	Automatically groups similar data points (machine learning)
8. Totals	Subtotals and grand totals in tables/crosstabs
9. Average Line	Line at the mean; compare each value with the average
10. Median with Quartiles	Median line with quartile band; central tendency and spread

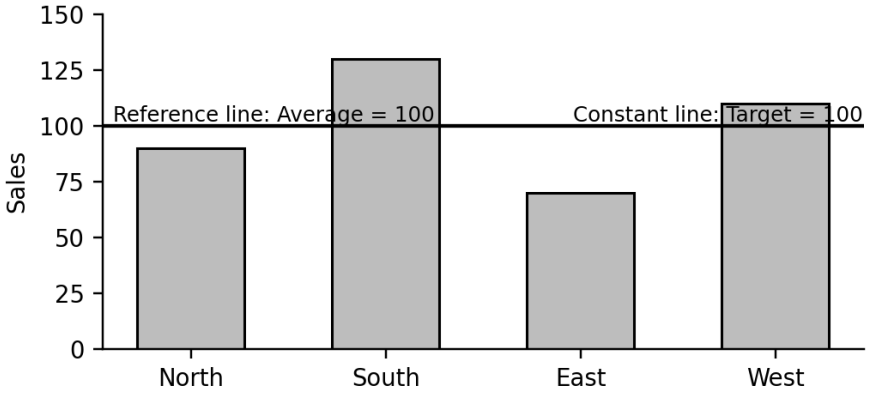


Figure: Constant line (fixed target) vs reference line (average of the data)

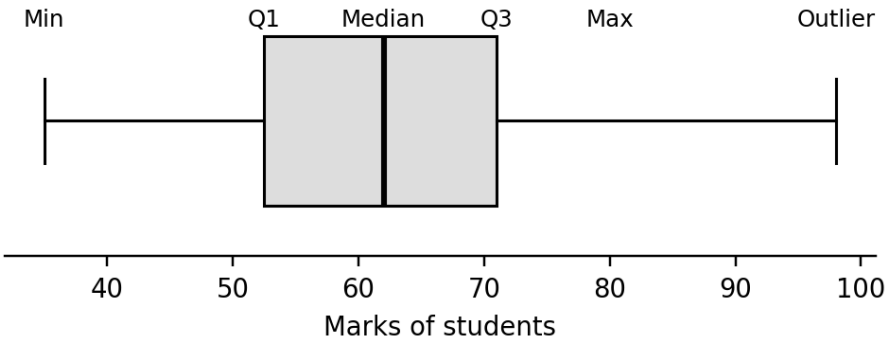


Figure: Box plot: Min, Q1, Median, Q3, Max and an outlier

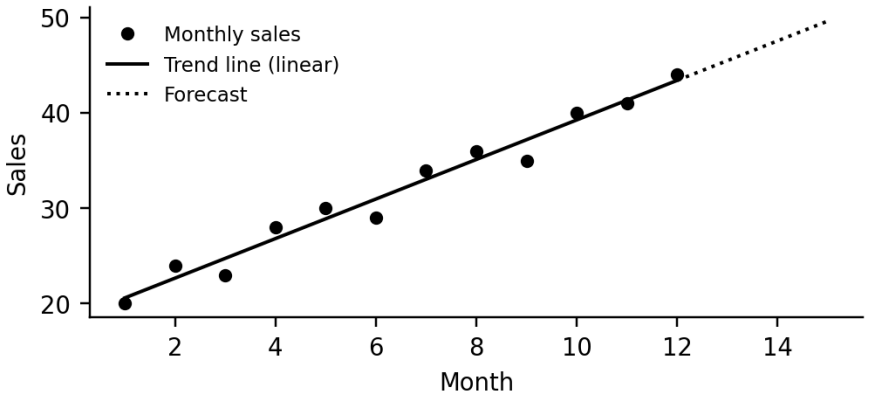


Figure: Trend line shows direction; forecast extends it into the future

How to use

1. Open a worksheet.
2. Click the **Analytics** tab beside the Data tab.
3. Select an object (e.g. Trend Line or Forecast).
4. Drag and drop it onto the view.
5. Choose where it applies: **Entire Table, Pane or Cell**.
6. Format and configure it.

Types of analysis supported

Type	Tools
Statistical	Average, Median, Standard deviation, Distribution, Quartiles
Predictive	Forecasting, Trend analysis, Time-series analysis
Comparative	Reference lines, Constant lines, Distribution bands
Pattern recognition	Clustering, Outlier detection, Trend identification

Benefits

Advanced analytics without coding; statistical insight; forecasting; finds trends, patterns and outliers; easy comparison with benchmarks; quick decisions; saves time; clearer dashboards.

Data Pane vs Analytics Pane

Feature	Data Pane	Analytics Pane
Purpose	Organizes and manages data fields	Adds analysis to visualizations
Contents	Dimensions, Measures, Parameters, Sets, Groups	Trend lines, Forecasts, Reference lines, Box plots, Clusters
Usage	Build visualizations	Enhance and analyse visualizations
Main function	Data preparation and field management	Statistical and predictive analysis
Interaction	Drag fields to Rows, Columns, Marks, Filters	Drag analytics objects onto the view

Applications

Sales analysis, financial reporting, business forecasting, customer behaviour, quality control, healthcare, education results, supply chain and inventory, marketing campaigns.

29. Canvas

The **Canvas** is the main visual area of a worksheet, dashboard or story where charts, tables, filters and other items are placed. In simple terms, **it is where data is turned into visual, interactive information**.

Key features

1. **Visualization area**: shows the chart.
2. **Drag-and-drop workspace**: drop fields from the Data Pane onto it.
3. **Marks**: data points shown as bars, circles, lines or shapes.
4. **Formatting**: colours, fonts, labels, borders.
5. **Interactive elements**: filters, parameters, tooltips and actions.
6. **Dashboard arrangement**: position and resize worksheets, filters, text and images.
7. **Multiple views**: combine worksheets into one dashboard.
8. **Sorting and filtering** of the data shown.
9. **Annotations**: highlight important points or trends.
10. **Responsive design**: adjust size/layout for different screens.

Canvas in different components

Component	Role of the Canvas
Worksheet	Where one visualization is built
Dashboard	Where many worksheets and objects are arranged

Component	Role of the Canvas
Story	Where a sequence of dashboards/views is presented
Map view	Where geographic data is shown on a map

30. Columns and Rows Shelves

The **Columns shelf** and **Rows shelf** define the **structure and layout** of a chart. Dragging fields onto them makes Tableau draw the chart automatically.

Columns shelf

- Controls what goes **across (horizontal)**.
- A dimension creates column headers; a measure creates a horizontal axis.
- Many fields can be placed; their order changes the layout.

Rows shelf

- Controls what goes **down (vertical)**.
- A dimension creates row headers; a measure creates a vertical axis.
- Many fields can be placed; their order changes the layout.

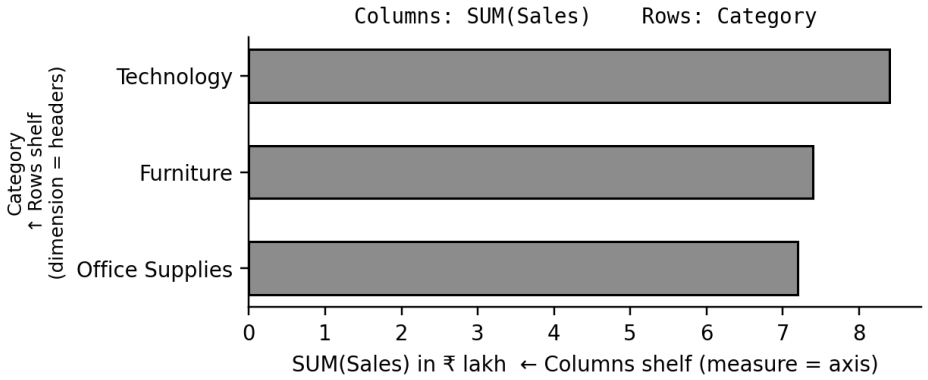


Figure: Sales on Columns + Category on Rows = horizontal bar chart

Examples

Rows shelf	Columns shelf	Result
Category	Sales	Bar chart of Sales by Category
Category	Profit	Bar chart of Profit by Category
Region	Sales	Sales by Region
Sales	Order Date	Sales trend over time (line chart)

Columns vs Rows

Columns shelf	Rows shelf
Horizontal arrangement	Vertical arrangement
Creates columns and horizontal axes	Creates rows and vertical axes
Fields shown left to right	Fields shown top to bottom
Compares categories across	Organizes categories down

Remember: *Columns = across the view. Rows = down the view.*

31. Filters and Pages Shelves

Filters shelf

The **Filters shelf** restricts the data shown in a chart so only data meeting certain conditions appears.

- Removes unwanted data.
- Filters by values, ranges, dates or conditions.
- Helps focus on relevant information.
- Can apply to one worksheet or many.
- A **filter card** lets users change the selection interactively.

Example: data has India, USA, UK, Australia. Put **Country** on Filters and select India; only Indian sales are shown.

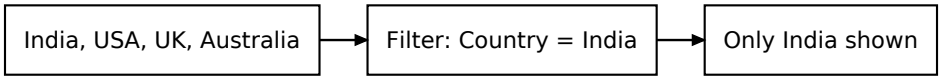


Figure: Filters shelf works like a sieve

Types of filters

Filter	Meaning	Example
Dimension filter	Filters categorical data	Region = East
Measure filter	Filters numeric values	Sales above ₹10,000
Date filter	Filters by dates / periods	Year 2024
Top N filter	Top or bottom N records	Top 10 products
Context filter	Runs first; other filters work within it	Region, then Top 10 inside it
Condition filter	Filters by a condition	Profit greater than 0

Pages shelf

The **Pages shelf** splits a chart into a **series of pages**, one for each value of a field. Controls: **Play, Pause, Previous, Next**. It works like an animation.

- Creates a sequence of views.
- Shows data one value at a time.
- Best for time-series analysis.
- Compares periods or categories.

Example: Year on Pages → 2019 → 2020 → 2021 → 2022 → 2023 → 2024. Press Play to watch the chart change.

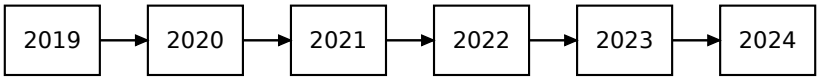


Figure: Pages shelf with Year: press Play to flip through years

Filters vs Pages

Filters shelf	Pages shelf
Restricts the data shown	Divides the view into sequential pages
Hides data that does not meet the condition	Shows each value one page at a time
For focused analysis	For sequential / time-based analysis
Filter card control	Play / Next / Previous controls

Filters shelf	Pages shelf
e.g. show only India	e.g. 2019, 2020, 2021... one by one

Remember: *Filters = choose what data you see. Pages = see the data one page at a time.*

Appendix: Important Differences at a Glance

Most 5-mark questions are 'differentiate between' questions. Revise these before the exam.

Compare	First	Second
Dimension vs Measure	Describes, categorical, blue, grouping	Numeric, green, aggregated
Fact vs Dimension	Measurable numbers, fact table	Descriptions, dimension table
Live vs Extract	Queries original source; real-time; needs connection	Copy in .hyper; faster; offline; needs refresh
Full vs Incremental refresh	Replaces the whole extract	Adds only new / changed rows
Join vs Union	Adds columns using a common field	Adds rows from similar tables
Join vs Relationship	Physical merge; can duplicate rows	Logical link; keeps level of detail
Join vs Blending	Same data source; row level	Different sources; aggregated level; primary/secondary
SQL vs NoSQL	Tables, fixed schema	Documents / key-value / graph / wide-column; flexible
Database vs Tableau data source	Stores data (DBMS)	Defines how Tableau uses the data
.twb vs .twbx	Workbook only (XML)	Workbook + data + images
.tds vs .tdsx	Connection only	Connection + extract + files
Logical vs Physical table	Related by relationships	Used for joins / unions inside a logical table
Menu Bar vs Toolbar	All commands, text menus	Frequent commands, icons
Data Pane vs Analytics Pane	Manage fields, build views	Add analysis to views
Columns vs Rows	Horizontal	Vertical
Filters vs Pages	Restricts data	Pages through data
Group vs Set	Merges values into one category	IN/OUT subset, can be dynamic
Keep Only vs Exclude	Shows only selected	Removes selected
Constant vs Reference line	Fixed value typed in	Calculated from data
Worksheet / Dashboard / Story	One chart	Many charts on one screen / a sequence of views